Sage Patag Lizarondo

Berkeley, CA | (619) 787-8282 | sage.lizarondo@berkeley.edu | linkedin.com/in/sagelizarondo

Education

Bachelor of Science in Mechanical Engineering **Expected Graduation December 2027**
University of California, Berkeley

Relevant Coursework: Manufacturing & Design Communication, Thermodynamics, Dynamic Systems and Feedback, Electronics for the Internet of Things, Heat Transfer

Engineering Experience and Projects

Chassis Team Member, UC Berkeley Solar Vehicle Team **Fall 2026 - Present**

- Designed a CFRP-aluminum honeycomb sandwich-panel lower occupant cell using a symmetric [0/45/core/45/0] layup of NT-350HT woven prepreg over a 19 mm 5052 Al core (2.80 kg/m², 6.40 kg total)
- Achieved a minimum factor of safety of 2.0 and peak deformation of 9.4 mm by simulating five 10 kN impact load cases (front, rear, three side positions) with Ansys composite analysis using Max Stress and Puck failure criteria
- Delivered a design 25% lighter than a 3-ply-per-side alternative (6.40 vs. 8.51 kg) while keeping minimum FoS at 2.0 or above

Productivity Device - *UC Berkeley Electronics Project* **Spring 2026**

- Modeled the enclosure in SolidWorks with an iPhone-sized cradle that aligns the phone over the force sensor, and created the system wiring diagram used for assembly
- Designed and built an ESP32-based study-focus device that detects when a phone is set down and starts a distraction-free session, sending push alerts over HTTP (ntfy) when the session is broken
- Integrated a 0.4–4 kg force-sensitive resistor for phone detection and 2 capacitive touch sensors for on/off control, with an amplified analog speaker playing 4 audio cues (on, off, ready, alarm)

Water Bottle Organizer - *UC Berkeley Manufacturing & Design Project* **Spring 2026**

- Designed and prototyped a modular clamp-on water-bottle organizer for phone, keys, and cards, iterating through multiple PLA FDM prototypes
- Specified 5 fits, including ASME B4.1 RC4 running and LC2 locational clearance fits for the hinge and dowel pins, a snap-fit latch, and a compliant bottle clamp, with GD&T per ASME Y14.5 on 3 functional datums

Wind Turbine Design - *UC Berkeley Three-Dimensional Modeling for Design Project* **Fall 2025**

- Designed and tested a 3-blade wind turbine and 16 in tower in SolidWorks under 17 in³ material and 9 × 9 in print-bed constraints; printed in ABS
- Developed low-Reynolds-number blades using an EYO9-8 airfoil, tapered chord (1.2 → 0.6 in), and non-uniform twist (18° → 4°), reaching 1.5 W peak at 4.75 V and 6,705 RPM in 25 mph wind
- Validated the tower with SolidWorks FEA (displacement, stress, factor of safety) and dial-indicator load testing to 1 kg, measuring 1.46 N/mm stiffness at 321 g tower mass

Bioinspired Rotors for Miniature UAVs - *UC Berkeley Experiments & Measurements Project* **(in progress)**

- Investigating whether bioinspired rotor geometries can make miniature UAV flight more energy-efficient or quieter through designed experiments

Leadership and Extracurricular Activities

Choreographer - *The 510 at UC Berkeley* **Fall 2026 - Present**

- Choreograph and teach original routines to 25 collegiate-level dancers for an end-of-semester showcase and smaller campus performances
- Design formations and blocking and lead twice-weekly rehearsals, giving individual feedback to prepare dancers for performances

Skills

Electronics & Programming: Python, MATLAB, ESP32 microcontrollers, sensor integration
CAD & Simulations: SolidWorks, Ansys Structural, SolidWorks Simulation (FEA)